



Automate & Manage Kubernetes add-ons' updates with GitOps

SRE France Meetup - February 2026

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1 - Context

1.1 - What is an "add-on"?

"Add-ons" are third party projects that provide functionality which extend the usability of Kubernetes.

Examples:

- [CoreDns](#)
- [Cert-Manager](#)
- [external-dns](#)
- [Ingress-nginx](#)
- [The Prometheus Operator](#)
- ...

Source: <https://kubernetes.io/docs/concepts/cluster-administration/addons/>

1 - Context

1.2 - A GitOps project example

```
src/
├── clusters/
│   ├── sdbx-team-1/
│   │   ├── test-feature--1/
│   │   └── test-temporal-1/
│   ├── staging/
│   │   └── test-project2-test/
│   └── tooling/
│       ├── prod-1/
│       └── test-1/
├── components/
│   ├── aws-ebs-csi-driver/
│   ├── aws-load-balancer-controller/
│   ├── aws-vpc-cni/
│   ├── cert-manager/
│   ├── cluster-autoscaler/
│   ├── external-dns/
│   ├── external-secrets/
│   └── kube-prometheus-stack/
└── ...
```

2 – Problematic

- **DRY** (Don't Repeat Yourself) branch
 - A single branch
 - Shared configuration across clusters
- **Multi-clusters**,
 - With a DRY layout, 1 PR = Big Blast radius
 - A single change can impact multiple environments
- **GitOps pattern**, merge then apply
 - Changes are applied after merge, not before
 - Rollback = Revert
 - Revert the changes on all clusters!

3 – IssueOps, Kesako?

3 - IssueOps, Kesako?

Atlantis



Since ~2015

Example change #5

[Open](#) Ikysow wants to merge 1 commit into `master` from `example`

Conversation 0

Commits 1

Files changed 1

 Ikysow commented just now Owner + 🗨️ ✎️

No description provided.

Example change

78e2acb

Add more commits by pushing to the `example` branch on `Ikysow/atlantis-example`.

 This branch has no conflicts with the base branch
Merging can be performed automatically.

[Merge pull request](#) You can also [open this in GitHub Desktop](#) or [view command line instructions](#).

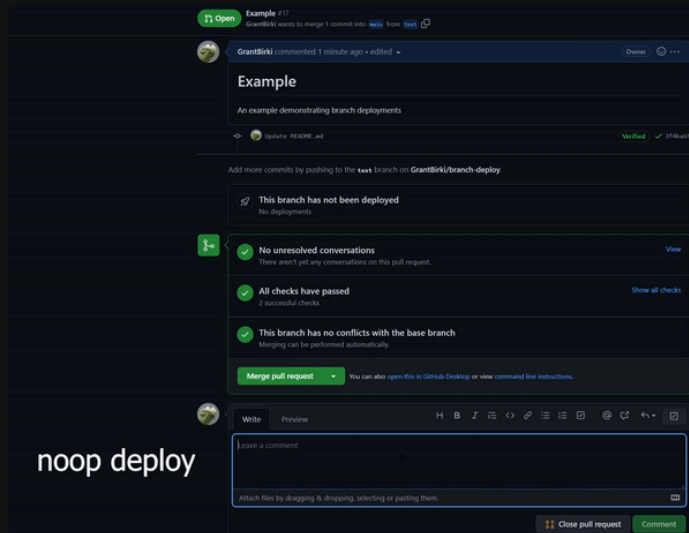
3 - IssueOps, Kesako?

IssueOps

- [Project](#)
- [github.blog](#)

"Like many of the other "Ops" tools, (ChatOps, GitOps, and so on), IssueOps leverages a friendly interface to drive behind-the-scenes automation. In this case, issues and pull requests (PRs) are the interface, and GitHub Actions is the automation engine."

—



noop deploy

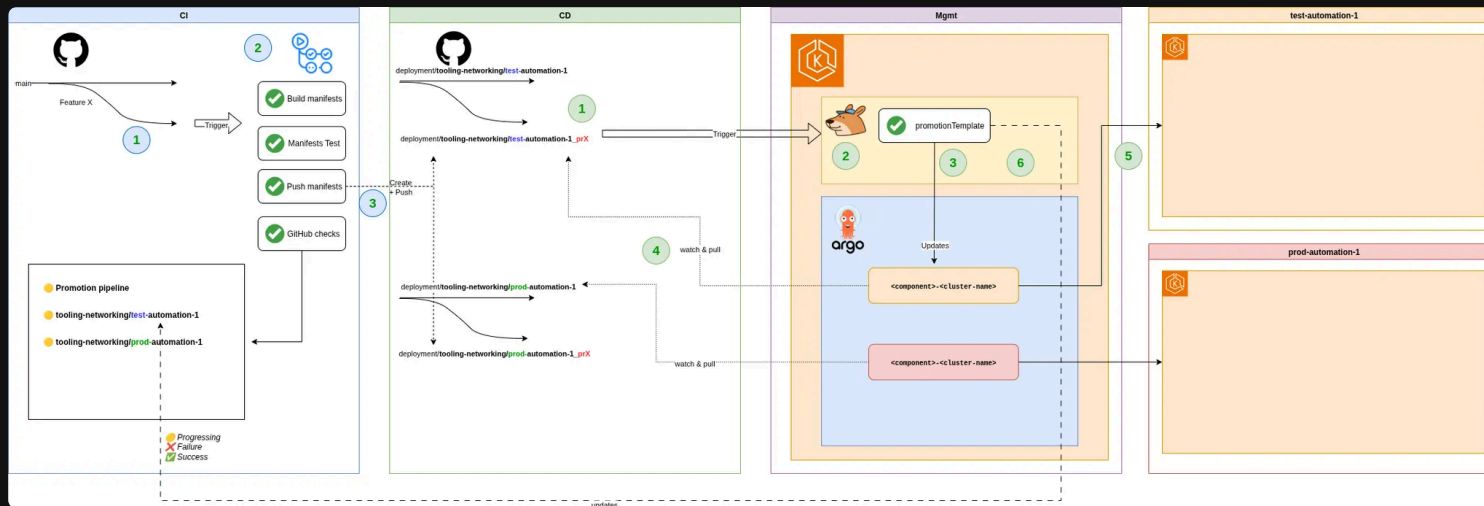
4 - Today's Project

Description

- 1 **GitHub repository** with the Kubernetes components configuration
- 2 kinds of **branches**:
 - 🏗️ **DRY** (Don't Repeat Yourself) branches
 - For features & human changes management
 - 💧 **WET** (Write Every Time) branches
 - For each EKS cluster, programmatically managed
- **Tooling:**
 - 🔄 **GitHub + GitHub Actions**
 - For the PR events, deployment API
 - [ArgoCD](#)
 - For the "GitOps" deployment
 - [Kargo.io](#)
 - For stages **promotion**
 - ✨ [github/branch-deploy](#)
 - For branch-based deployment

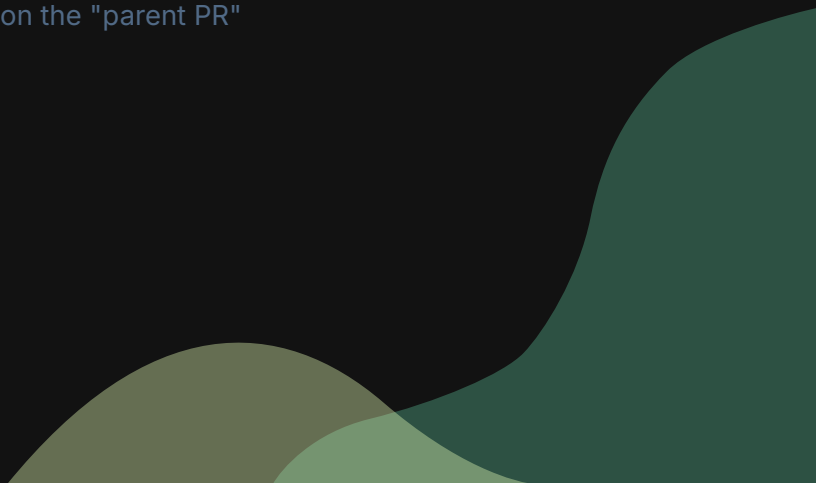
4.1 – Promotion pipeline

Overview



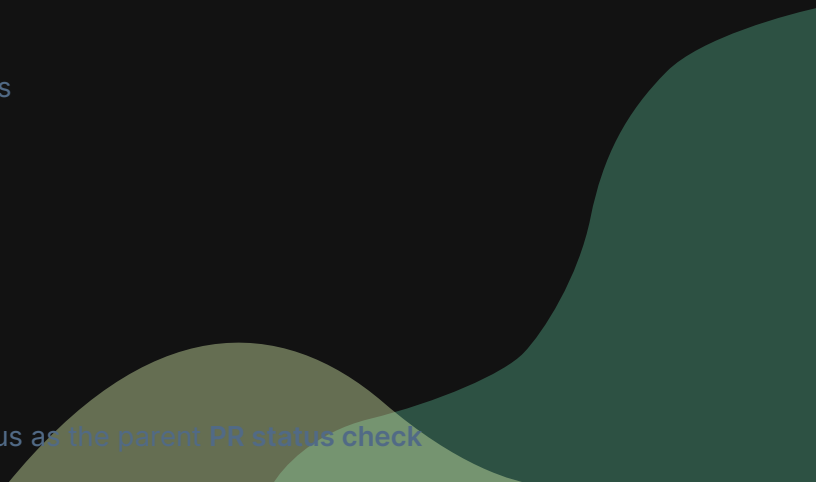
4.1 - Promotion pipeline - CI flow

1. A **feature PR** (or an automated one from dependencies management tools) is submitted
2. CI workflow is triggered
 - If the **changed files** impact multiple environments, according to their path,
 - "Hydrates" the Kubernetes manifests with `kustomize` + `helm`
 - Submits PRs against the **environment branches**
 - Sets status checks to "pending" for each of them on the "parent PR"



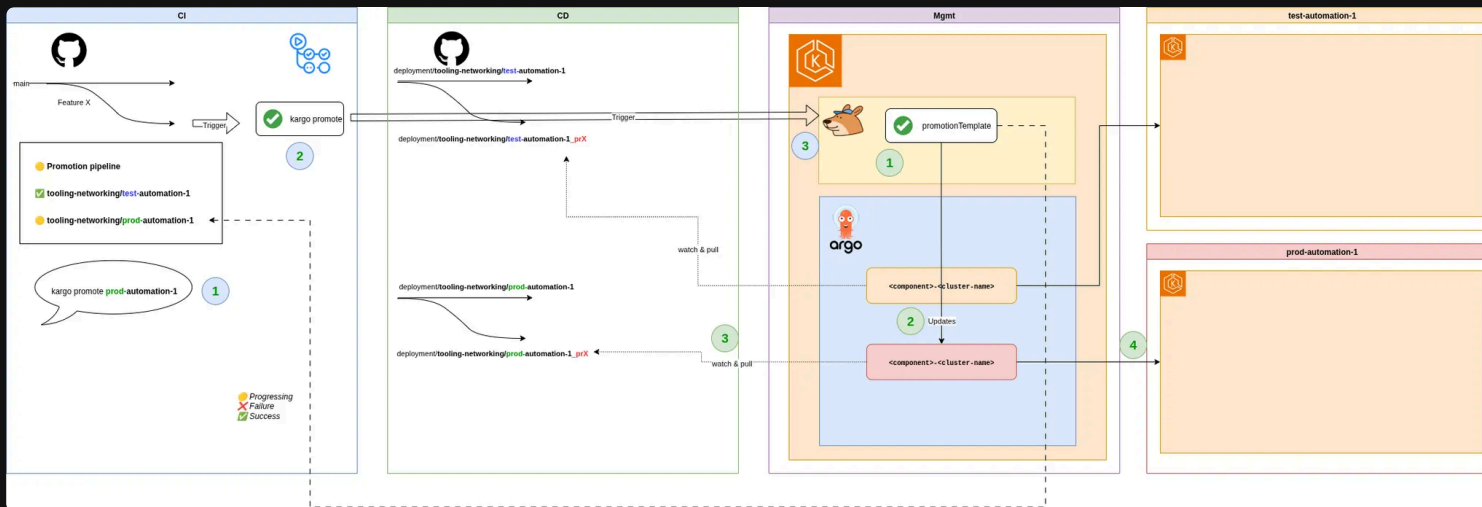
4.1 – Promotion pipeline – CD flow

1. Environment branches PR (+ git tags) trigger [Kargo](#)
2. The "testing" stage is triggered
3. Testing `PromotionTemplate` (aka workflow):
 - Updates the corresponding parent PR status check to "progressing"
 - Updates the target ArgoCD app to the Environment-branch's PR branch
 - Triggers an Application sync
 - Waits for ArgoCD to report the health and sync status
4. ArgoCD
 - Pulls the Environment-branch's PR branch
 - Deploys the changes to the target cluster
5. Kargo
 - Reports the ArgoCD Application health and sync status as the parent PR status check



4.2 – Stage promotion

Like Atlantis workflow, the PR must be approved & later actions are comments-driven.



4.2 – Stage promotion

CI flow:

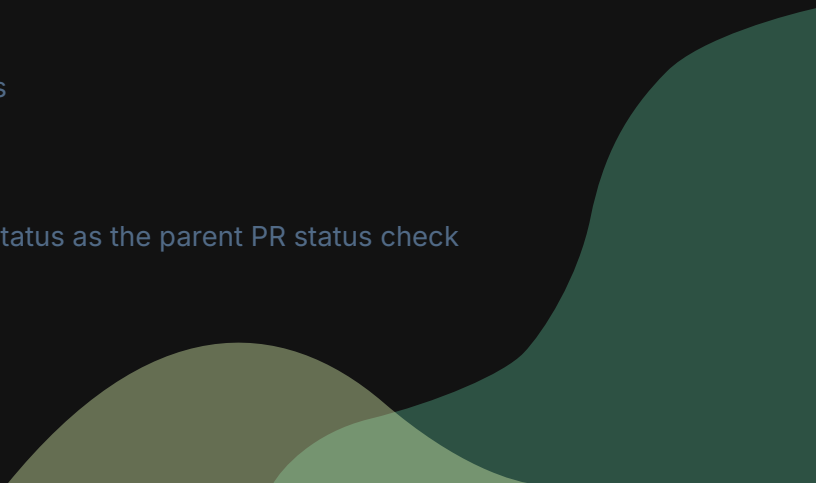
1. The operator can trigger the promotion with a **comment** on the Parent PR
2. The CI reacts to the comment and **triggers a Kargo promotion**



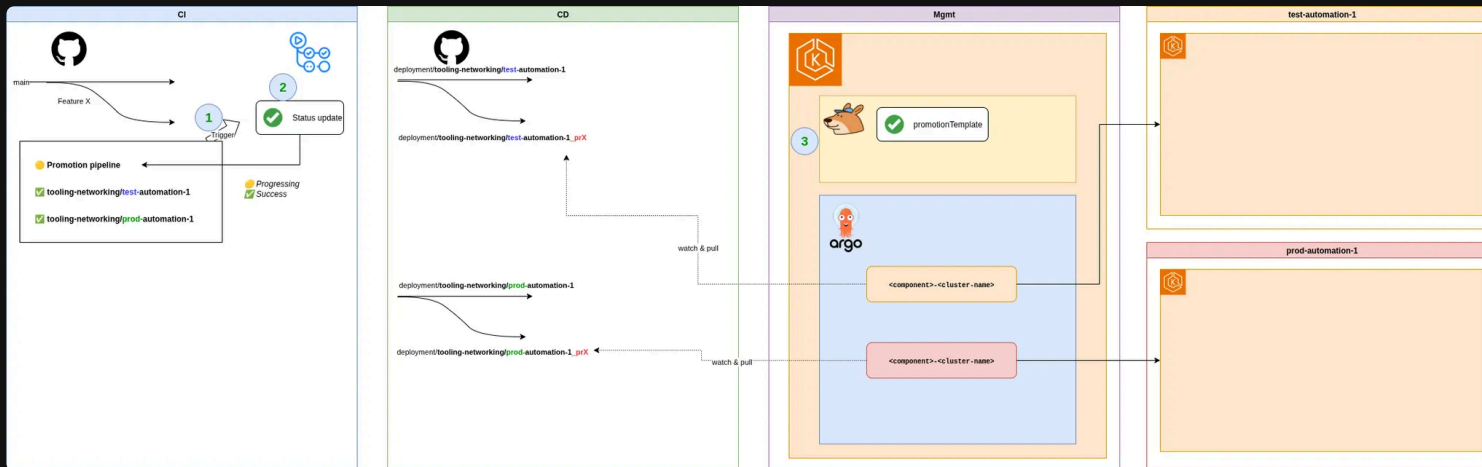
4.2 – Stage promotion

CD flow:

1. If the stage **promotion conditions** are met, Kargo runs the Production `PromotionTemplate`
2. The production `PromotionTemplate` :
 - Updates the corresponding parent **PR status check** to "progressing"
 - Updates the **target ArgoCD app** to the Environment-branch's PR branch
 - Triggers an Application **sync**
 - **Waits** for ArgoCD to report the health and sync status
3. ArgoCD pulls and deploys the changes
4. Kargo reports the ArgoCD Application health and sync status as the parent PR status check



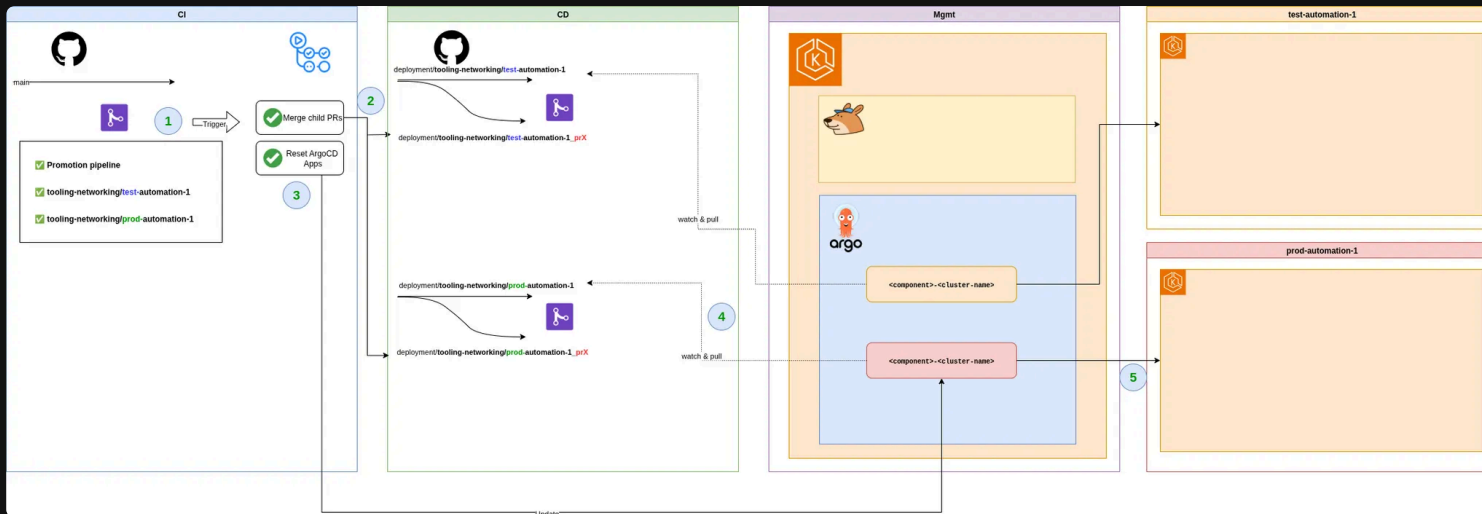
4.3 - PR "gating"



CI flow:

- A GitHub workflow can be triggered to react to **PR status checks updates**
- Updates the global **promotion pipeline** check according to the "child" checks

4.4 - Merge & Cleanup



4.4 – Merge & Cleanup

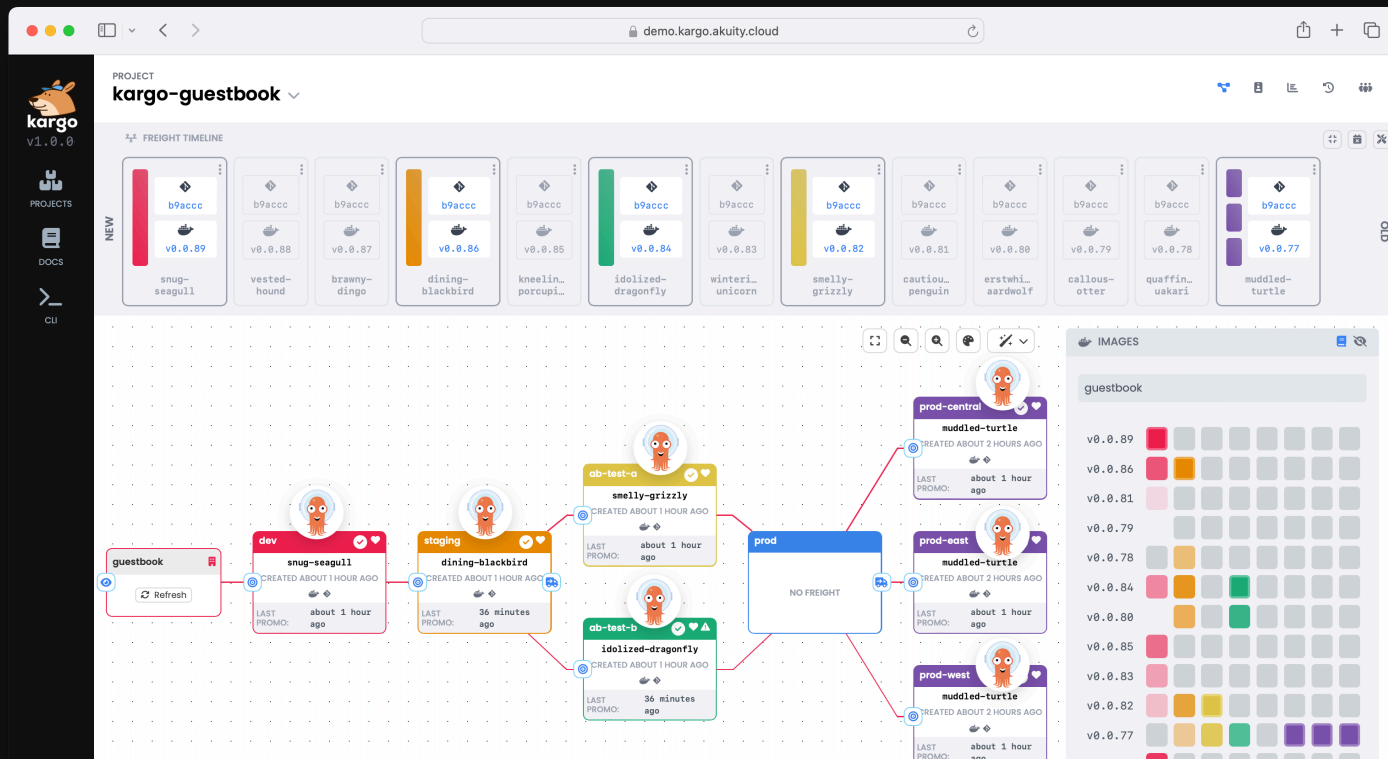
CI flow:

1. When the **parent PR** is merged
2. A CI Workflow is triggered
 - Gets the PR ID that has been merged
 - Gets child PRs from branch names
3. Merges the child PRs
4. Reset the ArgoCD Applications to the default Environment-branch



5 - Demo

5.1 - Demo - Kargo



5.1 - Demo - Kargo



Kargo Projects

```
---
apiVersion: kargo.akuity.io/v1alpha1
kind: ProjectConfig
metadata:
  # Name must match the project name
  name: kargo-cluster-components-tooling

spec:
  promotionPolicies:
    - stageSelector:
        matchLabels:
          environment: testing
        autoPromotionEnabled: true

    - stageSelector:
        matchLabels:
          environment: production
        autoPromotionEnabled: false
```

5.1 - Demo - Kargo

 Kargo Warehouse

```
---
apiVersion: kargo.akuity.io/v1alpha1
kind: Warehouse
metadata:
  name: cert-manager-pr
spec:
  subscriptions:
    - git:
        repoURL: https://github.com/DataDome/test-infra-eks-updates-p
        commitSelectionStrategy: NewestTag
        allowTags: ^cert-manager-pr-\d+$
```

5.1 - Demo - Kargo

Kargo Stages

```
1  ---
2  apiVersion: kargo.akuity.io/v1alpha1
3  kind: Stage
4
5  metadata:
6    name: cert-manager-production
7    labels:
8      environment: production
9    annotations:
10      kargo.akuity.io/color: "#B027F5"
11
12  spec:
13    requestedFreight:
14      - origin:
15          kind: Warehouse
16          name: cert-manager-pr
17        sources:
18          availabilityStrategy: All
19          stages:
20            - cert-manager-testing
```

5.1 - Demo - Kargo

Kargo Analysis Templates

An `AnalysisTemplate` is a resource that defines how to perform verification testing, including:

- Container images and commands to run
- Queries to external monitoring tools
- How to interpret results from metric providers
- Success or failure criteria
- Frequency and duration of measurements



5.1 - Demo - Kargo

Kargo Analysis Templates

`AnalysisTemplates` integrate natively with many popular open-source and commercial monitoring tools, including:

- Prometheus
- DataDog
- Amazon CloudWatch
- NewRelic
- InfluxDB
- Apache SkyWalking
- Graphite
- ...



5.2 - Demo - github/branch-deploy

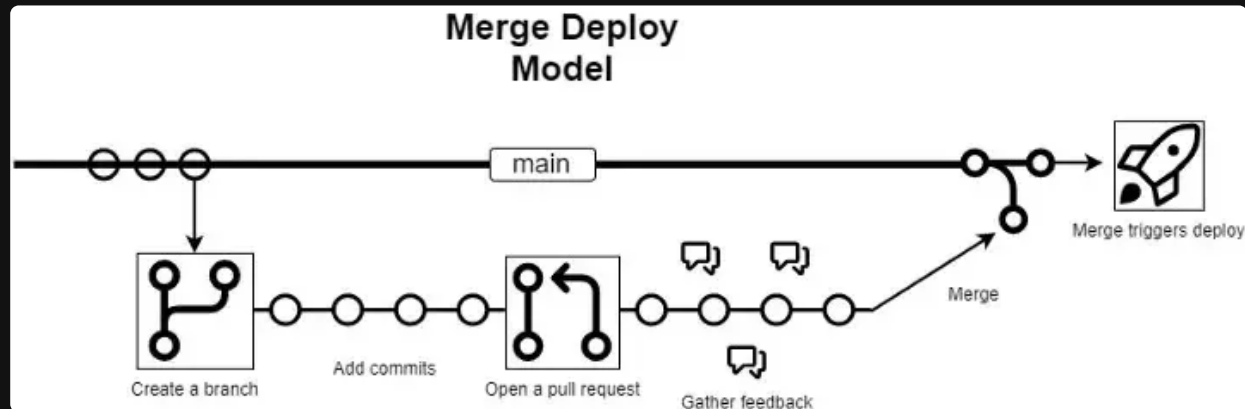
[github/branch-deploy](https://github.com/branch-deploy)



5.2 - Demo - github/branch-deploy

[github/branch-deploy](#)

Traditional flow



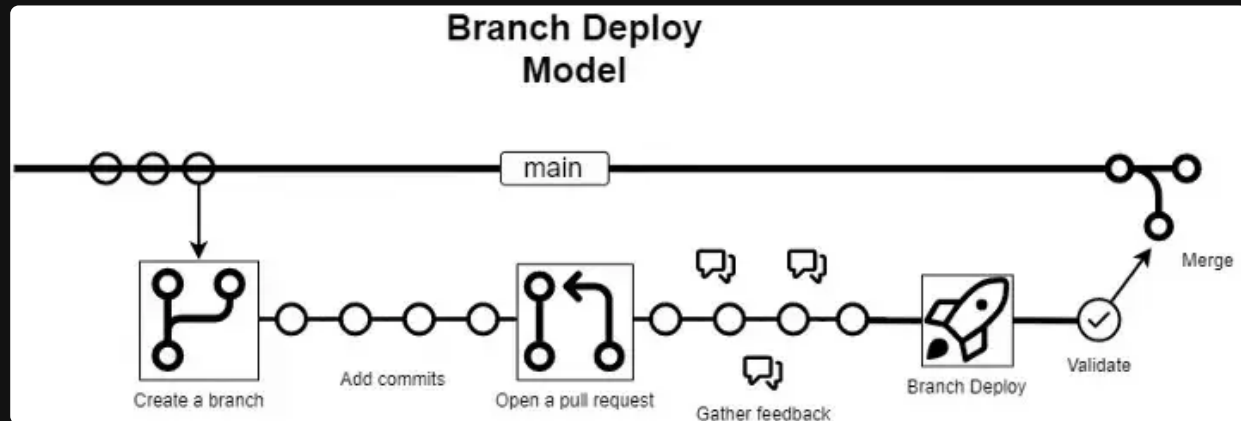
5.2 - Demo - github/branch-deploy

	COMMENT	DATE
○	CREATED MAIN LOOP & TIMING CONTROL	14 HOURS AGO
○	ENABLED CONFIG FILE PARSING	9 HOURS AGO
○	MISC BUGFIXES	5 HOURS AGO
○	CODE ADDITIONS/EDITS	4 HOURS AGO
○	MORE CODE	4 HOURS AGO
○	HERE HAVE CODE	4 HOURS AGO
○	AAAAA	3 HOURS AGO
○	ADKFJSLKDFJSDKLFJ	3 HOURS AGO
○	MY HANDS ARE TYPING WORDS	2 HOURS AGO
○	HAAAAAANDS	2 HOURS AGO

5.2 - Demo - github/branch-deploy

[github/branch-deploy](#)

New flow



5.2 - Demo - github/branch-deploy

[github/branch-deploy](#)

Features:

- 🔍 **Detects** when IssueOps commands are used on a pull request.
- 📄 **Configurable**: choose your command syntax, environment, noop trigger, base branch, reaction, and more.
- ✅ Respects your **branch protection settings** configured for the repository.
- 💬 **Comments** and reacts to your IssueOps commands.
- 🚀 Triggers **GitHub deployments** for you with simple configuration.
- 🌐 Configurable **environment** targets.
- 🔒 Deploy **locks** to prevent multiple deployments from clashing.



6 – Conclusion

Pros

- **Keeps** a “DRY” structure of the GitHub repository for human usage
- Uses **tools dedicated** to the “promotion” concerns
 - Kubernetes native
 - Promotion spec & policies
- **PR Driven flow**, like Atlantis,
 - Change-related central place for tracking and managing the deployments
- **Uses “WET” branches**,
 - Offloads manifests generation from ArgoCD
 - Allows checks of generated manifests (Lint, policies, ...)
 - Provides “clean” logs of changes (git log / audit-log) per cluster/environment
- **Only 1 new component is introduced**

6 – Conclusion

Cons

- The *Environment branches* and *Environment PRs* can feel overwhelming
- The PR status management might be “glue”, and therefore error-sensitive

Other projects

- <https://github.com/commercetools/telefonistka>
 - *A GitHub webhook server/Bot that facilitates change promotion across environments/failure domains in Infrastructure as Code(IaC) GitOps repos.*
- <https://gitops-promoter.readthedocs.io/en/latest/>
 - *Environment branches-based changes promotion*
 -  an ArgoProj labs project
- <https://github.com/projectsveltos>
 - *A Kubernetes Add-on Controller that Simplifies Add-on Management*

Other projects

Telefonistka

Other projects

ArgoProj – GitOps Promoter

Other projects

Sveltos

Thank you!

Questions?